Simone Shawn Cazzaniga

Design Engineer - DSP/Electroacoustics R&D

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I am committed towards crafting excellent audio and audio-adjacent algorithms and technology, all the way from concept to delivery. I constantly aim at the most impactful ways to best advance any team's work in the given timeframe, by fostering a cohesive collaborative effort. (Full résumé)

Italian@Native | English@C1 | Chinese@A1

PROFESSIONAL EXPERIENCE

Design Engineer - DSP/Electroacoustics R&D (BdSound) Milan **2023-08 - Current.**
I worked on core BdSound DSP/ML audio technology (BEV engine sound generation, echo and feedback cancellation, spatial audio and media processing) by prototyping algorithms in Python 3 and implementing them in C99 or C++20 (JUCE GUI). I built from scratch a Pybind11 compatibility layer to allow real-time use of C code from Python for research and quick demo purposes, increasing productivity for the whole team.

Technical Author (RISC-V Foundation) **2022-11 - 2023-03**
I wrote Chapters 3 through 6 of the RISC-V Fundamentals (LFD210) course.

Research Engineer Intern (ST Microelectronics) **2023-02 - 2023-07**

PUBLICATIONS

Effects of automotive microphone frequency response characteristics and noise conditions on speech and ASR quality - An experimental evaluation | AES Show '25 | DOI
Michele Buccoli, Yu Du, Jacob Soendergaard, Simone Shawn Cazzaniga **2025**

EDUCATION

M.Eng. Music&Acoustics Engineering (Politecnico di Milano) Milano (it) **110/110** **2022-12**
B.Eng. Automation&Control Engineering (Politecnico di Milano) Milano (it) **2020**
Diploma di Liceo Classico (Liceo Classico Bartolomeo Zucchi) Monza (it) **2016**

PROJECTS

IdPN Flutter Website **2019-09 - Current**
SHEAS - Simple Hardware Editor And Simulator **2022-07 - Current**
Introduction to Pipelining - CompArch Lesson Using SHEAS **2022**
Nand2Tetris Online JS Simulator Reimplementation **2021-03**
RISC-V RV32IMAC Sliderules **2021**

ONLINE & SINGLE COURSES

Numerical Modeling & Simulation in Acoustics (054274) **2024**
Science, Technology and Wikipedia PhD Course **2023-06 2023-07**
User Experience Research and Design Specialization (Auditor) **2025 - Current**
Nand2Tetris I&II (Auditor) **2020 2021**

VOLUNTEERING

RISC-V/LFX Mentor **2023**

1. **Bringing the "Sliderules" Cheat Sheets to the Web Mentorship (Summer 2023)**
2. **Add F-extension support to "Ripes" RISC-V Micro-Architectural Visual Educational Simulator (Autumn 2023)**

Camp Coordinator for S.C.I. camp IT-SCI 10.2 | Chivasso, Italy **2023-07**
Camp Coordinator for S.C.I. camp IT-SCI 13.1 | Rome, Italy **2022-08**
Local Event Coordinator for S.C.I. Workshops | Milan, Italy **2021 2024**
Volunteer for S.C.I. camp AM-HUJ 18.4 | Yerevan, Armenia **2019-07**
Volunteer for S.C.I. camp PT-BG 26.1 | Aldeia Nova, Portugal **2017-09**

C99 | C++20 | Python3 | NumPy | SciPy | Pandas | JUCE | VSCode | Git | Reaper | Jupyter